

Capacity Forecaster - Test Cases Document

localhost:5173

Database: SQLite Test Environment

Backend URL: http://localhost:8000

Frontend URL: http://localhost:5173

AI Service: Google Gemini API

Test Case ID	Title	Objective	Status
TC-01	Backend Health Check	Verify backend is running successfully.	Pass / Fail
TC-02	Login with Valid Credentials	Verify successful login functionality.	Pass / Fail
TC-03	Login with Invalid Password	Verify login validation.	Pass / Fail
TC-04	Register New User	Verify user registration.	Pass / Fail
TC-05	Reject Weak Password	Validate password security rules.	Pass / Fail
TC-06	Dashboard Summary Load	Verify dashboard data loading.	Pass / Fail
TC-07	Forecast Disk Threshold	Verify disk forecasting functionality.	Pass / Fail
TC-08	Forecast CPU Threshold	Verify CPU forecasting functionality.	Pass / Fail
TC-09	Forecast Memory Threshold	Verify memory forecasting functionality.	Pass / Fail
TC-10	Invalid Forecast Query	Verify fallback behavior when AI fails.	Pass / Fail
TC-11	Quick Scan	Verify quick scan functionality.	Pass / Fail
TC-12	Valid CSV Upload	Verify CSV upload functionality.	Pass / Fail
TC-13	Invalid File Upload	Verify CSV validation.	Pass / Fail
TC-14	Insufficient Dataset	Verify dataset row	Pass / Fail

	Validation	validation.	
TC-15	Forecast History Save	Verify forecast history functionality.	Pass / Fail
TC-16	Restore History	Verify restoring previous forecasts.	Pass / Fail
TC-17	Delete Single History Item	Verify deleting a saved forecast.	Pass / Fail
TC-18	Clear History	Verify clearing all saved history.	Pass / Fail
TC-19	Update Forecast Settings	Verify settings persistence.	Pass / Fail
TC-20	Update Alert Thresholds	Verify threshold configuration.	Pass / Fail
TC-21	Toggle Theme	Verify dark/light mode functionality.	Pass / Fail
TC-22	Update Profile Name	Verify profile update functionality.	Pass / Fail
TC-23	Change Password	Verify password update functionality.	Pass / Fail
TC-24	Reject Wrong Current Password	Verify password validation.	Pass / Fail
TC-25	Logout	Verify logout functionality.	Pass / Fail

Acceptance Criteria

Backend starts successfully, frontend starts successfully, authentication works, dashboard loads metrics, forecast generation works, CSV upload works, history works, settings persist, and logout functions correctly.